

# Jaiveer Bassi

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## Experience

### Physical Intelligence

May 2026 – Present

*AI Robot Operator*

*San Francisco, CA*

- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through manipulation, sorting, and assembly tasks.

### Stanford University

Sep 2026 – Present

*Stanford SSG Fellow*

*Stanford, CA*

- Conduct mentor-guided independent research by developing a core curiosity question, evaluating primary scientific literature, and formulating testable hypotheses.

### Handshake AI

Apr 2026 – Jun 2026

*Contracted AI Response Evaluator*

*Remote*

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

### Castline Studio

Jan 2026 – Apr 2026

*Operator & Developer – Resin 3D Printing, Next.js, TypeScript*

*Vallejo, CA*

- Operated resin 3D-printing production for an e-commerce business generating \$2K+/month across 100+ product listings, including calibration, curing, post-processing, and tolerance control.
- Built an order-management platform with client-side STL parsing for automated geometry, volume, dimension, and print-cost calculations.

### Kaiser Permanente Regional Laboratory

Oct 2023 – Jun 2024

*Software Developer*

*Berkeley, CA*

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

## Selected Projects

### CAN Bus Fault-Injection & Diagnostics Rig | C++, Arduino, MCP2515 (SPI), SocketCAN

- Assembled a three-node CAN network of Arduino microcontrollers driving MCP2515 CAN controllers over SPI, with a USB-CAN interface and a custom wiring harness, then induced bus-off, arbitration, connector, wiring, and termination faults.
- Mapped physical failure modes to reproducible SocketCAN error states and multimeter evidence in a published diagnostic catalog.

### Scribbot Desktop Drawing Arm | C++, Arduino Nano, PCA9685 (I2C), OpenSCAD

- Designed in OpenSCAD and built a 3D-printed four-servo drawing arm controlled by an Arduino Nano driving a PCA9685 PWM controller over I2C.

### GELLO-Style Leader-Follower Teleoperation Rig | Arduino, ADC Sensing, Servo Control

- Built a four-DOF leader-follower mechanism for approximately \$30 by assembling linkages, wiring potentiometer-based joint encoders, and calibrating direct joint-space control without inverse kinematics.

### $\pi$ 0 on a Budget | OpenPI, LoRA, LeRobot, JAX, Arduino

- Built a synchronized data pipeline recording 30 FPS video and joint commands from a custom 4-DOF teleoperation arm, with conversion into a LeRobot-compatible dataset.
- Implemented an RTX 4070 workflow for  $\pi$ 0-FAST fine-tuning, memory profiling, policy serving, closed-loop evaluation, and a real-time dashboard of live Arduino telemetry and predicted action chunks.

### UR5e Fault Injection & Diagnostics Harness | RTDE, URSim, Fault Injection

- Built a control and fault-characterization harness that drives simulated robot motion and logs synchronized joint telemetry while testing connection failures, stale handles, hangs, and native crashes.

Research Publications

When Does INT8 Actually Accelerate Host-CPU Inference?

- Audited MobileNetV2 INT8 conversion and packaging, showing that INT8 latency depends on backend delegation and quantization granularity; v1.2.0 archived on Zenodo.

One Byte, One Rank Reversal: Auditing HumanEval Pipeline Sensitivity

- In a pinned five-model RTX 4070 reproduction, StarCoder2-3B scored 3/164 against its published 31.7%; removing one trailing prompt newline raised it to 49/164 and restored the ranking.

Not All Bad Demonstrations Are Equally Bad

- Ran 660 controlled policy fits across five demonstration failure modes, two policy families, ten seeds, and transition-matched controls; under review at the CoRL 2026 LfC workshop.

Sampling Budget Biases Apparent Metastable-State Counts in Molecular Simulation

- Used controlled synthetic landscapes and two independent alanine-dipeptide trajectories to show how sampling budget can inflate apparent state counts; under review at JCTC.

Service

Reviewer, CoRL 2026 Workshop on Learning from Corrections and Interventions (LfC) 2026

Technical Skills

**Languages:** C++, Python, C#, TypeScript, JavaScript, SQL, Bash  
**Embedded & Hardware:** Arduino, microcontrollers, SPI, I2C, CAN/SocketCAN, PWM servo and actuator control, sensor wiring, PCB rework, soldering, multimeter diagnostics  
**Robotics & Systems:** RTDE, URSim, ROS2, real-time control, fault injection, telemetry analysis, Linux, Git, Docker, Kubernetes, PostgreSQL  
**Machine Learning:** PyTorch, JAX, LeRobot, OpenPI, LoRA, ONNX, NumPy, model evaluation, CUDA

Education

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Grand Canyon University</b><br><i>M.S. in Computer Software Engineering</i> <ul style="list-style-type: none"><li>• Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages.</li></ul> | May 2024 – Oct 2025<br>Phoenix, AZ  |
| <b>University of Silicon Valley</b><br><i>B.S. in Computer Science</i> <ul style="list-style-type: none"><li>• Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine for real-time playback.</li></ul>                                                           | May 2021 – Aug 2023<br>San Jose, CA |

Certifications

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|-----------------------------------------------------------|-------------------------------------------------------|
| Accelerated Computing in Modern CUDA C++ (Sep 2026)       | HackerRank Software Engineer Certificate (Mar 2026)   |
| Cisco Network Support and Security (Sep 2025)             | IBM Artificial Intelligence Fundamentals (Aug 2025)   |
| Stanford AI and Machine Learning in Healthcare (Jun 2025) | Google IT Support Professional Certificate (May 2021) |